

Ten Open Source Cloud Computing Resources You Should Know About

Cloud computing is the preferred choice for most organisations today, and a number of open source products are available in this domain. We take a look at some of the best.



Cloud server management, cloud management and cloud monitoring products are the basic ingredients for the perfect cloud recipe. Open source initiatives for IaaS, cloud monitoring and cloud management have gained popularity not only in private cloud implementations, but their use is now being pioneered in hybrid cloud implementations too. Cloud management and monitoring services are ideal for checking on the performance of resources, compliance auditing and security. The elastic and flexible nature of the cloud has been extensively used by the open source software framework that supports data-intensive distributed applications. Let's look at some of the popular open source cloud computing resources.

1. OpenStack

OpenStack offers open source software to build public and private clouds. It was started by Rackspace (the Cloud Files platform) and NASA (Nebula) in 2010. Currently, Intel, Cisco, Dell, HP, AMD, Canonical, SUSE Linux, Red Hat and IBM have joined the OpenStack project. It is released under the Apache

licence. There are three main service families under OpenStack: 1) The Presentation Layer interacts with users to accept and present information; 2) the Logic Layer handles intelligence and control functionalities for the cloud; and 3) the Resources Layer contains compute, storage and network resources for the cloud.

OpenStack has a modular architecture, which includes components such as compute, object storage, image service, identity, dashboard, networking, and block-storage. OpenStack has Amazon EC2 and Amazon S3-compatible APIs.

The use cases are:

- Service providers offering an IaaS compute platform.
- IT departments provisioning compute resources to teams and projects.
- Big Data processing with Hadoop.
- Scaling up and scaling down resources to meet demand for Web resources and applications.

2. CloudStack

CloudStack is an open source cloud computing software for creating, managing, and deploying Infrastructure as a Service